

Annotated Bibliography

Note: Annotations are provided as an aid to the reader and are the personal opinions of the author. No endorsements of the works or ideas presented are implied.

Categories:

- Overviews, Fact & Fiction
- Introductions to Emerging Physics
- Detailed Technical Papers

Overviews, Fact & Fiction:

1. Mallove, E.F., and Matloff, G.L., *The Starflight Handbook*, Wiley Science Editions, John Wiley & Sons, Inc., New York (1989).

Book; (274 pages) **An overview of interstellar travel**, the problems and proposed solutions. It contains technical details sufficient to lead researchers to more substantive material, yet is easy enough to read for general interest. The book constrains itself mostly to existing science and projected technology.

2. Mauldin, J.H., *Prospects for Interstellar Travel*, American Astronautical Society by Univelt Inc. San Diego CA, (1992).

Book; (370 pages) **An overview of interstellar travel**, the problems, proposed solutions and social issues. It contains sufficient details to lead researchers to more substantive material. It also touches on longer-term and more speculative ideas, including space warps, Zero Point Energy and Higgs fields.

3. Emme, E.D., (ed), *Science Fiction and Space Futures Past & Present*, American Astronautical Society Historical Series, Vol 5, (1982). Book; Explores the **connection between Science Fiction and advances in astronautics**. Conclusions: Science fiction is not technically accurate at predicting the future, but has been instrumental in inspiring famous pioneers. Examples are cited.

4 Hirsch, D. & Zimmerman, H., *Starlog Photo Guidebook to Spaceships (Enlarged Edition)*, Starlog Press, O'Quinn Studios, NY NY, 1980, 2nd printing, ISBN 0-931064-23- 6, (1982)

Book; (98 pages); Chronicles the cinematic and television art of science fiction space vehicles from their beginning through the publication date. Photos of the vehicles and brief descriptions of the movies or series from which they came are provided.

5. Herbert, N., *Faster Than Light, Superluminal Loopholes in Physics*, Plume Books, New York (1988,89).

Book; Presents the variety of evidence and speculation toward faster-than-light possibilities.

6. Hujsak, John T, & Hujsak Edward, **Interstellar Propulsion Society**

A non-profit organization for professional and public membership that provides a means for scientists and engineers, worldwide, to join in collaborative efforts to accelerate scientific and engineering

advancements in space propulsion, leading to manned missions to other star systems at fractional light speeds, relativistic velocities and beyond. Its World Wide Web Internet address is:
<http://www.digimark.net/ips/>.

7. Kurtis,B., producer, "**The Science of Star Trek**"; a video episode of "**THE NEW EXPLORERS**"; series, Kurtis Productions, Ltd. and WTTW Chicago, aired January 1995. One hour video comparing the science fiction of the popular series "Star Trek"; to the science fact of today.

Introductions to Emerging Physics:

8. Bennett, G., "**Warp Speed, Fact or Fiction?**", In *Final Frontier*, p. 35-39, (September-October 1994) Article (5 page); Discusses the issues and unanswered questions of Faster-Than-Light phenomena; causality issues, time travel implications, tachyon theory requiring imaginary mass, and wormhole theory. It mentions the conclusions of the May 1994 workshop at NASA JPL on this topic.

9. Szpir, M, "**Spacetime Hypersurfing?**", In *American Scientist*, Vol. 82, p. 422-423, (September-October 1994) Article (2 page); Explanation of the 1994 Alcubierre "warp drive"; article, including a copy of the figure of the warped spacetime from Alcubierre's paper. Discusses the necessity for the existence of negative energy density and the relation of that issue to the Casimir effect.

10. Gonick, L., "**Science Classics (Warp-and-woof drive)**", In *Discover*, p. 44- 54, (DEC 1994). Comic-Strip (2 page); Playful and succinct explanation of the 1994 Alcubierre "warp drive"; article.

11. Peterson, I., "**A New Gravity?, Challenging Einstein's General Theory of Relativity**", In *Science News*, Vol. 146, p. 376-378, (Dec 3, 1994).

Article; Introduces the theory of Yilmaz and Alley that suggest that corrections are required to the Einstein field equations in order for them to correctly predict real behavior. The article also introduces experiments planned by NASA Goddard to test the theory by measuring the light propagation velocities over two different paths.

12. Matthews, R., "**Inertia: Does Empty Space Put Up the Resistance?**", In *Science*, Vol. 263, p. 612-613, (4 Feb 1994).

Article (2 pages); An introduction to the theory of Haisch, et al. (Phys Rev A, Feb 94), where inertia is described as a high-frequency electromagnetic drag relative to the Zero Point Fluctuations of the vacuum. An important note is that this article mentions an experiment that would have a bearing on this theory: Exposing high-energy electrons to a terawatt beam from a neodymium-YAG laser, an experiment planned by K.McDonald et. al.@ Stanford Linear Accelerator Center, for late 1994.

13. Clarke, A. C., "**Space Drive: A Fantasy That Could Become Reality**", In *Ad Astra*, p. 38, (Nov-Dec 1994)

Article (1 page); Arthur C. Clarke suggests that the time has come to revisit the idea of the "mythical space drive"; based on the recent theories of Haisch, Rueda and Puthoff (Phys Rev A, Feb 94).

14. Von Baeyer,H.C., "**Vacuum Matters**", In *Discover*, Vol. 13, No. 3, p. 108- 112, (March 1992). Article: Description of vacuum properties, including a historical perspective. Describes Zero Point Energy, Virtual Pairs, and the Casimir effect. Introduces recent experiments by D.Kleppner of MIT and

N.Lawandy and J.Martorell of Brown University where a Casimir cavity is used to inhibit the spontaneous emission of exited atoms.

15. Boyer,T.H., "The Classical Vacuum", In *Scientific American*, p. 70-78, (Aug 1985).

Article (10 pg): Summary of the contents of the vacuum: Thermal radiation; Zero Point Energy; Casimir Effect; and the ZPE as viewed during constant and accelerated motion.

16. Muller,R.A., "The Cosmic Background Radiation and the new Aether Drift", In *Scientific American*, Vol. 238, N. 5, p. 64-74, (May 1978).

Article; Introduction to the CBR. "The three-degree cosmic background radiation provides an all-pervasive radiation 'aether' ..." (it also mentions Mach's Principle) ... against which we can measure our velocity. (Earth/Sun are moving about 300km around our galaxy which itself is moving 600 km/sec relative to the universe. The universe is rotating less than 10^{-9} arc-sec/century) There is some discussion of the expansion of the universe and the slight non-isotropic nature of the CBR. Most of the article deals with the difficulties and methods of measuring the CBR.

17. Krauss, L. M., "Dark Matter in the Universe", In *Scientific American*, p. 58-68, (Dec 1986).

Article; Describes evidence for and the theories leading to the conclusion that there must be more matter in the universe than is visible. Estimates range that only 2%-30% of the mass is visible. The missing mass is called "dark matter."; Unconfirmed candidates for this missing mass are the neutrino and the following forms of "cold dark matter": axions, photinos, and cosmic strings. There is no mention of ZPE or virtual pairs as candidate explanations. The evidence: observed rotation rate of spiral galaxies is too high; Hubble's constant and the assumption of a "flat" universe requires more mass than is observed; Virial Theorem and observations of the relative velocities of stellar masses for stable systems requires more mass than is observed, and nucleosynthesis which predicts proportions between light elements and heavier elements requires more mass than is observed. The assumption that the universe is flat is based on the isotropy of the CBR and theories that predict that a non-flat universe would be obvious.

18. Hough,J., (Dept. of Natural Philosophy, Univ. of Glasgow, Glasgow G12 8QQ, UK), "New Detectors for Gravity Waves", In *Nature*, Vol. 316, p. 576-577, (15 AUG 1985).

Article (2 pg); Clear explanation of a variety of gravitational wave detectors, who is doing what, and speculates on sources of gravitational radiation.

19. Pool,R., "Closing in On Einstein's Special Theory of Relativity", In *Science*, Vol. 250, p. 1207-1208, (Nov. 9, 1990).

Article; Summarizes a variety of recent positive tests of relativity theory.

Detailed Technical Papers:

20. Alcubierre, M., (Dept. of Physics and Astronomy, Univ. of Wales, College of Cardiff CF1 3YB, UK), "The warp drive: hyper-fast travel within general relativity", In *Classical and Quantum Gravity*, Vol 11, p. L73-L77, (1994).

Letter to the editor (5 pg); Quoting from the abstract: "It is shown how, within the framework of general relativity and without the introduction of wormholes, it is possible to modify space-time in a way that

allows a spaceship to travel with an arbitrarily large speed. By a purely local expansion of space-time behind the spaceship and an opposite contraction in front of it, motion faster than the speed of light as seen by observers outside the disturbed region is possible. The resulting distortion is reminiscent of the "warp drive"; of science fiction. However, just as happens with wormholes, exotic matter will be needed in order to generate a distortion of space-time like the one discussed here." It is unknown in physics whether such "exotic matter"; having a negative mass or negative energy density can exist. Classical physics tends toward a "no," while quantum physics leans to a "maybe, yes." Equal amounts of positive energy density matter will also be required. There is presently no known way to induce or control such effects. It is also uncertain whether this whole "warp"; would indeed move faster than the speed of light. Even though light-speed is a limit *within* spacetime, the rate at which spacetime itself can expand or contract is an open issue. Back during the early moments of the Big Bang, spacetime is assumed to have been able to expand faster than the speed of light. This is known as the "inflationary universe" perspective.

21. Haisch, B. (Lockheed Palo Alto CA 94304), Rueda, A. (Dept. of E.E. California State Univ. Long Beach CA 90840) & Puthoff, H.E. (Inst. for Advanced Studies at Austin TX 78759), "**Inertia as a Zero-Point Field Lorentz Force**";, In *Physical Review A*, Vol. 49, No. 2, p. 678-694, (FEB 1994)

Article: Asserts that inertia, the resistance to a change in velocity, is a high-frequency electromagnetic drag relative to the Zero Point Fluctuations of the vacuum (there is also a lower frequency drag effect called the Davies-Unruh effect, 1976). Inertial mass is calculated and compared to the gravitational mass calculated from Puthoff's 1989 paper that states that gravity is a consequence of ZPF interactions. They are off by a factor of two which is discussed. Unfortunately, this perspective does yet not provide clues for how to experimentally manipulate inertia or gravity using electromagnetics. The paper offers no examples of calculating mass from this perspective, such as calculating the mass of a neutron. Theoretically, the possible weak links of the theory are the reliance on "parton"; perspectives (assuming that all matter is fundamentally constructed of charged particles) and the Abraham-Lorentz radiation damping equation.

22. Puthoff, H.E., (Institute for Advanced Studies at Austin, Austin TX 78746), "**Gravity as a zero-point-fluctuation force**", In *Physical Review A*, Vol. 39, N. 5, (A89- 33278), p. 2333-2342, (Mar 1, 1989).

Article; Describes gravity as a side-effect of the Zero Point Fluctuation effects on charged matter. It assumes that all matter is fundamentally constructed of charged particles. Specifically it equates gravity to Van der Waals forces resulting from the dipoles induced by ZPE oscillations. Newton's $G = \frac{5}{12} \frac{(\hbar \times \text{cut-off frequency} \sim 10^{42} \text{ Hz})}{m^2}$ Unfortunately, this perspective does not yet provide clues for how to experimentally manipulate gravity using electromagnetics.

23. Yilmaz, H. (Hamamatsu Photonics, K. K. Hamamatsu City, 435 Japan, Electro- Optics Technology Center, Tufts Univ. Medford Mass, 02155 USA), "**Toward a Field Theory of Gravitation**", In *Il Nuovo Cimento*, Vol. 107B, no. 8, p. 941-960, (1991?).

Article; Suggests that corrections are required to the Einstein field equations in order to correctly predict behavior for systems beyond the simple one-body problem (the case of a single gravitating body affecting an insignificant test particle). The required correction is to add a term for the gravitational field energy itself to the matter tensor. This implies that the gravitational field has a mass-energy equivalence itself.

24. Misner, C.W., Thorne, K.S., and Wheeler, J.A., *Gravitation*, W.H. Freeman & Co., New York (1973). Book; (1279 pages) This is a **primary text book on General Relativity** and related physics.

25. *Physics Through the 1990's: Gravitation, Cosmology, and Cosmic Ray Physics*, PB 86-241486, National Research Council, Washington DC, for the Department of Defense, (ISBN 0-309-03579-1), (1986). Report (187 pages): Easy to understand compilation of the "State Of the Art"; of experimental work on the subject and recommends for future work, including candidate space experiments.

26. Forward, R.L., (Forward Unlimited, Malibu CA 90265-7783), "**Space Warps: A Review of One Form of Propulsionless Transport**", AIAA-89- 2332, 25th Joint Propulsion Conference, Monterey CA, (1989). Paper; Examines a variety of ideas for creating worm holes, space warps or other propulsive methods based on theories from General Relativity. None of the methods can be practically accomplished.

27. Winterberg, F., (Desert Research Institute, Univ of Nevada System, Reno, Nevada), "**On Negative Mass Propulsion**", AIF Paper 89-668, 40th Congress of the International Astronautical Federation, Malaga, Spain, (Oct, 1989).. Article; A propulsion scheme is described consisting of negative and normal mass side-by-side. Because of the attraction / repulsion properties of the respective masses, both masses accelerate continuously in the direction pointing from the negative to the positive mass. Suggests that there is experimental evidence for the existence of negative mass. It also discusses an older interpretation of relativity by Lorentz and Poincar, that relies on an absolute reference frame; ether. This treatment is consistent with experimental observations including the Michelson-Morley experiment. The null result of the Michelson-Morley experiment is explained that the apparatus is equally affected as the light, canceling any observed effect. It also discussed the role of ZPE and mentions Zitterbewegung in the context of experimental evidence for the existence of negative mass.

28. Forward, R.L., (Forward Unlimited, Malibu CA 90265-7783), "**Negative Matter Propulsion**", AIAA-88-3168 preprint, 24th Joint Propulsion Conference, Boston, Mass., (1988). {Note: a shorter version that omits the gravitational coupling details exists as: Forward, R.L., "Negative Matter Propulsion", In *AIAA Journal of Propulsion*, Vol. 6 No. 1, p. 28- 37, (Jan-Feb 1990)} Paper (19 pg); Shows that the negative mass propulsion scheme does NOT violate conservation of momentum even though no reaction mass is expelled. Three cases are analyzed classically; where the masses are gravitationally coupled, coupled with an ideal spring, and coupled by electrostatic forces. Each of these couplings is analyzed assuming the magnitudes of the negative and positive masses are equal, and again assuming they are unequal. For all situations, energy and momentum are found to be conserved. It is important to note that in the case of gravitational coupling, the analysis is simplified by setting the initial velocities such that their momentums sum to zero. This zero sum continues throughout the different resulting motions of each mass. For the case of spring-coupled unequal masses, oscillations occur and energy is conserved using the potential energy of the spring.

29. Millis, M.G., (NASA, Lewis Research Center), "**Exploring the Notion of Space Coupling Propulsion**", In *Vision 21: Space Travel for the Next Millennium*, Symposium Proceedings, Apr 1990, NASA-CP-10059, p. 307- 316, (1990).

Paper; Identifies the 3 major objections to the notion of propelling a vehicle by coupling to the properties of space: (1) violates conservation of momentum, (2) no reactive media in space, (3) no grand unified theories to link gravity with other controllable phenomena. Introduces a variety of avenues to possibly satisfy these objections, including introducing the notion of coupling to distant masses via the fundamental structure of the intervening space and the associated need for alternative treatments of space physics.

30. Bennett, G.L. & Knowles, H.B., (NASA HQ & Physics consulting, El SoBrante, CA 94803),

"Boundary Conditions on Faster-Than-Light Transport Systems", AIAA paper 93-1995, presented at the 29th Joint Propulsion Conference, Monterey CA, (1993). Paper (16 pg); Outlines the prerequisite physics to enable Faster-Than-Light transport: The theory of FTL must equally describe sub-light physics, must not violate causality, must match physics at quantum scales, and must satisfy conservation of energy. The paper discusses Tachyons and Wormholes concluding that it is "very unlikely that interactive tachyons exist";, and the we don't know enough about Wormholes to render judgement.

31. Forward, R.L., (Hughes Research Lab, Malibu CA 90265) "**Feasibility of Interstellar Travel: A Review**", In *Acta Astronautica*, Vol. 14, p 243-252, (1986)

Article: Comments on a variety of scientifically plausible propulsion schemes. States that Centauri cluster is reachable within 50 yrs given a velocity of 10% lightspeed, and 25 stars are within reach given 30% lightspeed. Various propulsion schemes are compared, but the comparisons were not on equal footing:

- Nuclear Electric Propulsion: 10,000 year trip
- Nuclear (Fission) Pulse Propulsion: 130 year trip.
- Antimatter Propulsion: 50 year mission requires 9 kg of anti-hydrogen for 1 ton payload.
- Nuclear Fusion: To be determined.
- Interstellar Ramjet: To be determined, particularly the issue of fusing protons without having to slow them down.
- Beamed Power: 20 year trip for a crewed vehicle, but needs lasers that are about 10,000 times more powerful than all the power used on the Earth today.



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Due to the large number of correspondences that Marc Millis now receives, he regrets that he might not be able to answer more detailed inquiries at this time.

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